

Git Tutorial: Push Your First Project to GitHub

This guide assumes you already created an empty repository on GitHub.

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1. Initialize Git

```
git init
```

Creates a new Git repository in your current project folder.

2. Stage All Files

```
git add .
```

Stages every new and modified file, preparing them for the next commit.

3. Connect to Your GitHub Repository

Replace `<repository-url>` with your GitHub repository URL.

```
git remote add origin <repository-url>
```

Example:

```
git remote add origin https://github.com/username/my-project.git
```

4. Create Your First Commit

Use a clear, descriptive commit message instead of vague messages like "update" or "fix".

```
git commit -m "Initial project setup"
```

Good First Commit Messages

```
git commit -m "Initial project setup"
git commit -m "Create project structure"
git commit -m "Implement landing page"
git commit -m "Add authentication system"
git commit -m "Configure backend API"
git commit -m "Integrate database connection"
git commit -m "Complete portfolio website"
git commit -m "Launch version 1.0"
```

A good commit message explains what changed, not just that something changed.

5. Push to GitHub

If your default branch is `master`:

```
git push -u origin master --force
```

If your default branch is `main` (recommended for new repositories):

```
git push -u origin main --force
```

What the Flags Mean

- `-u` — Sets the upstream branch so future pushes only require `git push`.
- `origin` — The remote repository name.

- `main` / `master` — The branch to push.
- `--force` — Forcefully replaces the remote branch with your local branch.

Use `--force` carefully. It overwrites the remote branch history and can remove commits that already exist on GitHub. For most first-time pushes to an empty repository, it isn't necessary. A regular push is usually enough:

```
git push -u origin main
```

Complete Workflow

```
git init

git add .

git remote add origin https://github.com/username/repository.git

git commit -m "Initial project setup"

git push -u origin main
```

If your repository uses `master` instead of `main`:

```
git push -u origin master
```

This is the standard workflow for publishing a new project to GitHub.